PRAKTIKUM BASIS DATA

MODUL 3

Scripting_SQL_Basic

Oleh:

Revalsa Putra Lusyandra 103122430011

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

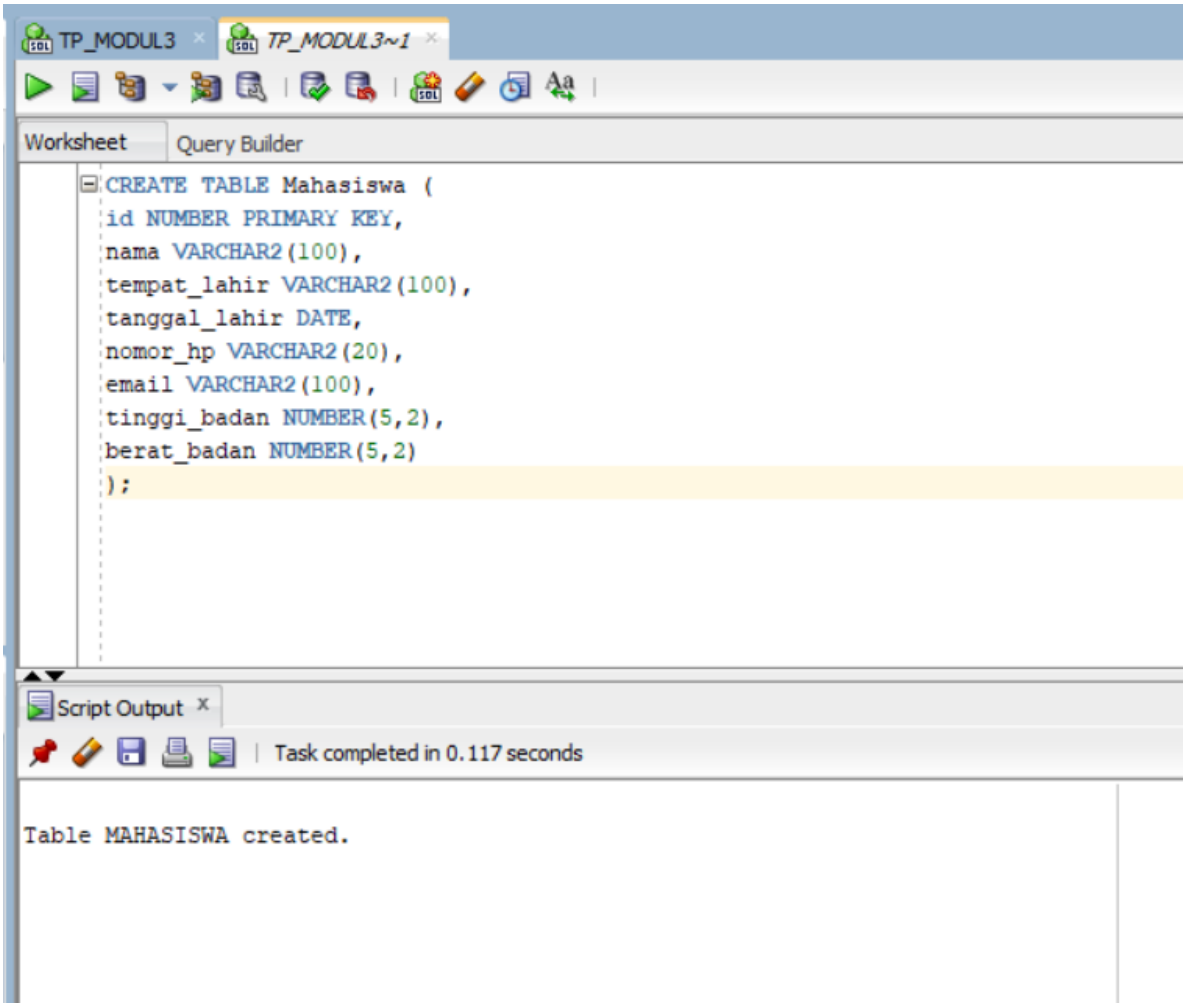
DIREKTORAT KAMPUS PURWOKERTO

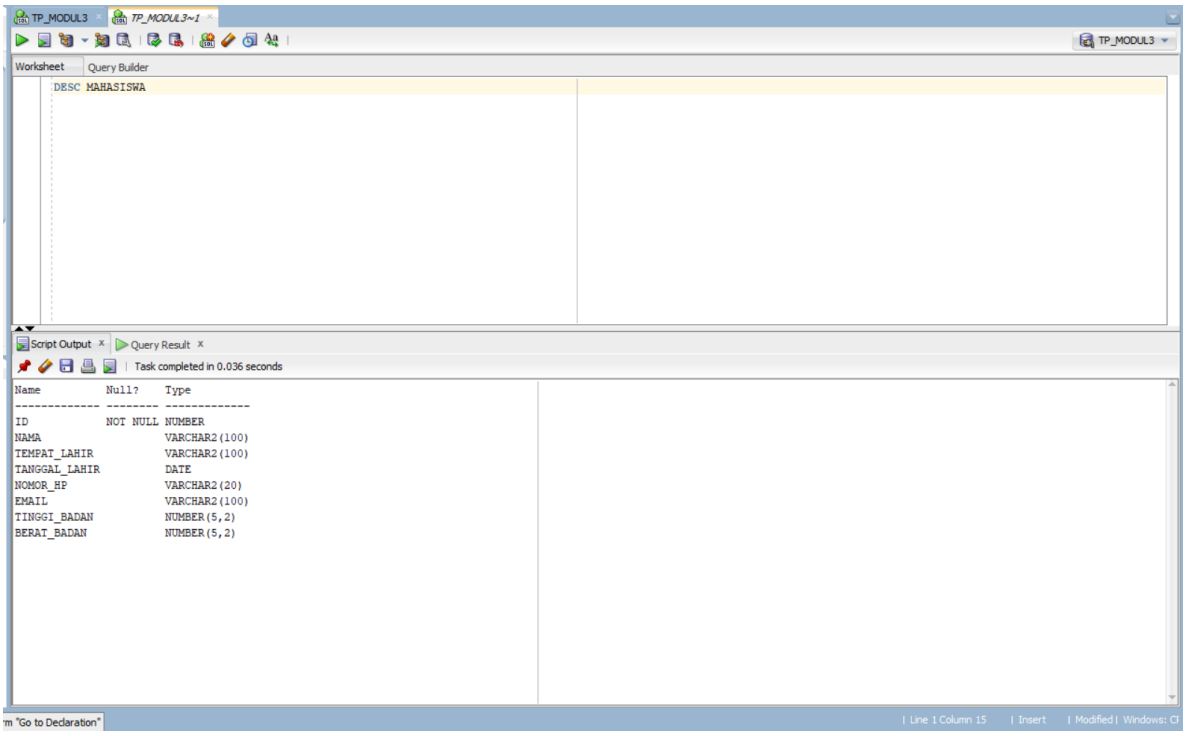
UNIVERSITAS TELKOM

2026

TP

1. Buatlah sebuah table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.




TP_MODUL3
TP_MODUL3~1
TP_MODUL3
Worksheet
Query Builder
DESC MAHASISWA
Script Output
Query Result
Task completed in 0.036 seconds
Name Null? Type
ID NOT NULL NUMBER
NAMA VARCHAR2(100)
TEMPAT_LAHIR VARCHAR2(100)
TANGGAL_LAHIR DATE
NOMOR_HP VARCHAR2(20)
EMAIL VARCHAR2(100)
TINGGI_BADAN NUMBER(5,2)
BERAT_BADAN NUMBER(5,2)
"Go to Declaration"
Line 1 Column 15
Insert
Modified
Windows: CI

2. Lakukan insert data pada table nomor 1 sebanyak 15 rows.

```
INSERT ALL
INTO Mahasiswa VALUES (1,'Revalsa Putra Lusyandra','Banjarnegara',TO_DATE('2006-05-04','YYYY-MM-DD'),'085640075728','vallreey@gmail.com',170,60)

INTO Mahasiswa VALUES (2,'Cheval Grand','Jakarta',TO_DATE('2006-02-15','YYYY-MM-DD'),'0856234567002','cheval@gmail.com',168,65)

INTO Mahasiswa VALUES (3,'Symboli Rudolf','Jakarta',TO_DATE('2000-03-20','YYYY-MM-DD'),'085664567003','kaichou@gmail.com',160,50)

INTO Mahasiswa VALUES (4,'Takumi Fujiwara','Dieng',TO_DATE('2001-04-12','YYYY-MM-DD'),'085634567004','fujiwara@gmail.com',172,68)

INTO Mahasiswa VALUES (5,'Manhattan','Semarang',TO_DATE('2007-05-18','YYYY-MM-DD'),'0856234567005','manhattan@gmail.com',158,48)

INTO Mahasiswa VALUES (6,'Agnes Tachyon','Medan',TO_DATE('2006-06-25','YYYY-MM-DD'),'085234567006','tachyon@gmail.com',175,70)

INTO Mahasiswa VALUES (7,'Satsuki Koto','Malang',TO_DATE('2006-07-07','YYYY-MM-DD'),'085534567007','kotosatsuki@gmail.com',162,52)

INTO Mahasiswa VALUES (8,'Ren Senjyu','Bandung',TO_DATE('2001-08-30','YYYY-MM-DD'),'085534567008','nexusblue@gmail.com',174,66)

INTO Mahasiswa VALUES (9,'Kira Yamato','Padang',TO_DATE('2002-09-09','YYYY-MM-DD'),'085734567009','freedom@gmail.com',159,49)

INTO Mahasiswa VALUES (10,'Pria Solo','Solo',TO_DATE('2001-10-11','YYYY-MM-DD'),'085634567010','joko@gmail.com',171,67)

INTO Mahasiswa VALUES (11,'Admire Vega','Bogor',TO_DATE('2006-11-03','YYYY-MM-DD'),'085684567011','ayabe@gmail.com',157,47)

INTO Mahasiswa VALUES (12,'Matikanefukukitaru','Makassar',TO_DATE('2006-12-14','YYYY-MM-DD'),'085534567012','fukuku@gmail.com',173,69)

INTO Mahasiswa VALUES (13,'Mayano Top Gun','Bali',TO_DATE('2007-01-21','YYYY-MM-DD'),'085234567013','maya@gmail.com',161,53)

INTO Mahasiswa VALUES (14,'Ryosuke Takahashi','Palembang',TO_DATE('2001-02-28','YYYY-MM-DD'),'085634567014','rx7fc@gmail.com',169,64)

INTO Mahasiswa VALUES (15,'Kadoya Tsukasa','Pontianak',TO_DATE('2002-03-17','YYYY-MM-DD'),'086534567015','magentta@gmail.com',176,72)

SELECT * FROM dual;
```

Script Output | Query Result

Task completed in 0.041 seconds

```
15 rows inserted.
```

TP_MODUL3 | TP_MODUL3~1

Worksheet | Query Builder

```
SELECT * FROM Mahasiswa
```

Script Output | Query Result

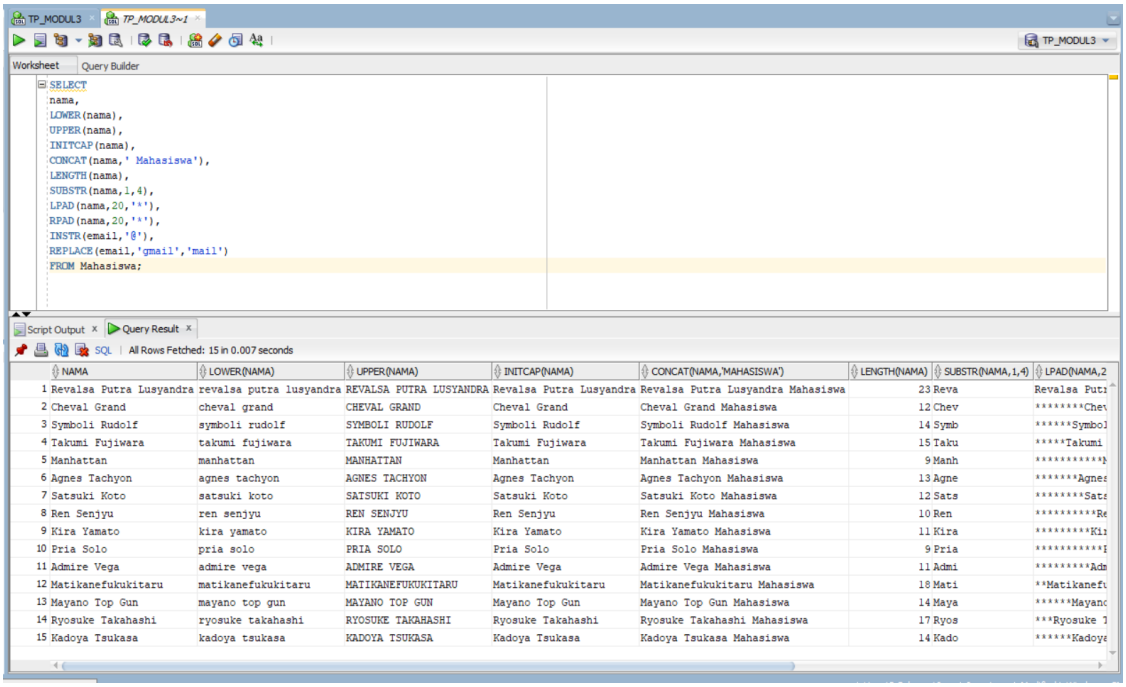
SQL | All Rows Fetched: 15 in 0.006 seconds

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NOMOR_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN
1	1	Revalsa Put...	Banjarnegara	04-MAY-06	085640075728	vallreey@gmail.com	170	60
2	2	Cheval Grand	Jakarta	15-FEB-06	085623456...	cheval@gmail.com	168	65
3	3	Symboli Rudolf	Jakarta	20-MAR-00	085664567003	kaichou@gmail.com	160	50
4	4	Takumi Fuji...	Dieng	12-APR-01	085634567004	fujiwara@gmail.com	172	68
5	5	Manhattan	Semarang	18-MAY-07	085623456...	manhattan@gmail...	158	48
6	6	Agnes Tachyon	Medan	25-JUN-06	085234567006	tachyon@gmail.com	175	70
7	7	Satsuki Koto	Malang	07-JUL-06	085534567007	kotosatsuki@gma...	162	52
8	8	Ren Senjyu	Bandung	30-AUG-01	085534567008	nexusblue@gmail...	174	66
9	9	Kira Yamato	Padang	09-SEP-02	085734567009	freedom@gmail.com	159	49
10	10	Pria Solo	Solo	11-OCT-01	085634567010	joko@gmail.com	171	67
11	11	Admire Vega	Bogor	03-NOV-06	085684567011	ayabe@gmail.com	157	47
12	12	Matikanefuk...	Makassar	14-DEC-06	085534567012	fukuku@gmail.com	173	69
13	13	Mayano Top Gun	Bali	21-JAN-07	085234567013	maya@gmail.com	161	53
14	14	Ryosuke Tak...	Palembang	28-FEB-01	085634567014	rx7fc@gmail.com	169	64
15	15	Kadoya Tsukasa	Pontianak	17-MAR-02	086534567015	magentta@gmail.com	176	72

m "Go to Declaration" | Line 1 Column 24 | Inse

3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:

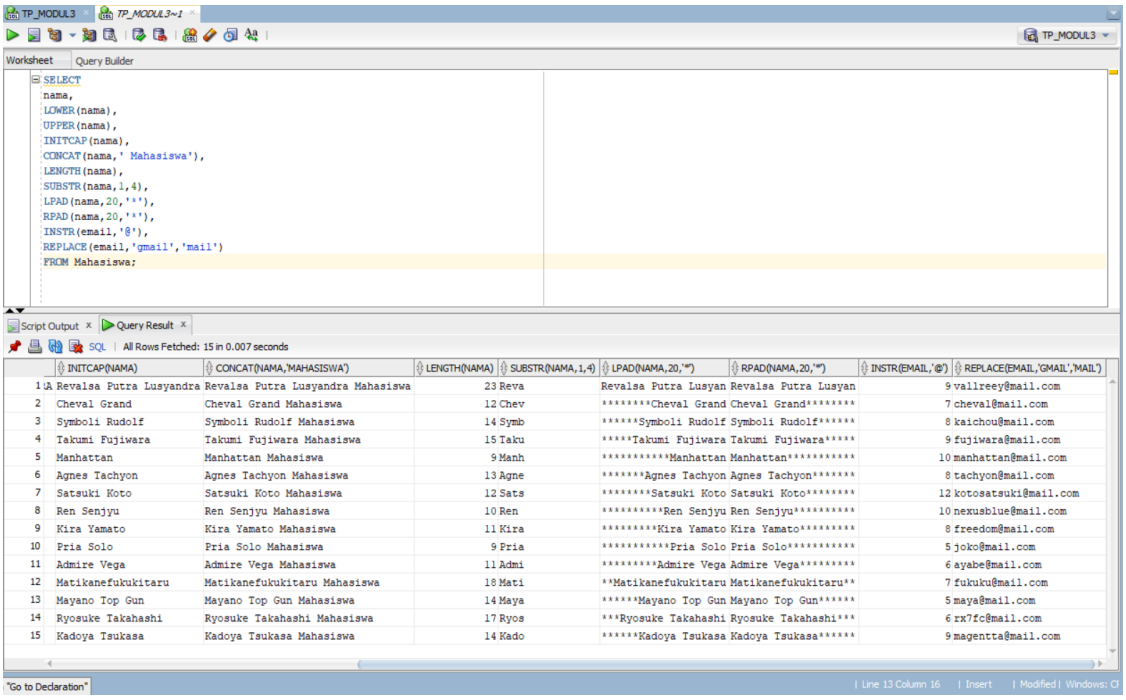
   a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)



```
SELECT
nama,
LOWER(nama),
UPPER(nama),
INITCAP(nama),
CONCAT(nama,' Mahasiswa'),
LENGTH(nama),
SUBSTR(nama,1,4),
LPAD(nama,20,'*'),
RPAD(nama,20,'*'),
INSTR(email,'@'),
REPLACE(email,'gmail','mail')
FROM Mahasiswa;
```

All Rows Fetched: 15 in 0.007 seconds

	NAMA	LOWER(NAMA)	UPPER(NAMA)	INITCAP(NAMA)	CONCAT(NAMA,'MAHASISWA')	LENGTH(NAMA)	SUBSTR(NAMA,1,4)	LPAD(NAMA,2
1	Revalsa Putra Lusyandra	revalsa putra lusyandra	REVALSA PUTRA LUSYANDRA	Revalsa Putra Lusyandra	Revalsa Putra Lusyandra Mahasiswa	23	Reva	Revalsa Put
2	Cheval Grand	cheval grand	CHEVAL GRAND	Cheval Grand	Cheval Grand Mahasiswa	12	Chev	********Chev
3	Symboli Rudolf	symboli rudolf	SYMBOLI RUDOLF	Symboli Rudolf	Symboli Rudolf Mahasiswa	14	Symb	******Symbol
4	Takumi Fujiwara	takumi fujiwara	TAKUMI FUJIWARA	Takumi Fujiwara	Takumi Fujiwara Mahasiswa	15	Taku	*****Takumi
5	Manhattan	manhattan	MANHATTAN	Manhattan	Manhattan Mahasiswa	9	Manh	***********M
6	Agnes Tachyon	agnes tachyon	AGNES TACHYON	Agnes Tachyon	Agnes Tachyon Mahasiswa	13	Agne	*******Agnes
7	Satsuki Koto	satsuki koto	SATSUKI KOTO	Satsuki Koto	Satsuki Koto Mahasiswa	12	Sats	********Sats
8	Ren Senjyu	ren senjyu	REN SENJYU	Ren Senjyu	Ren Senjyu Mahasiswa	10	Ren	**********Re
9	Kira Yamato	kira yamato	KIRA YAMATO	Kira Yamato	Kira Yamato Mahasiswa	11	Kira	*********Kir
10	Pria Solo	pria solo	PRIA SOLO	Pria Solo	Pria Solo Mahasiswa	9	Pria	***********P
11	Admire Vega	admire vega	ADMIRE VEGA	Admire Vega	Admire Vega Mahasiswa	11	Admi	*********Adm
12	Matikanefukukitaru	matikanefukukitaru	MATIKANEFUKUKITARU	Matikanefukukitaru	Matikanefukukitaru Mahasiswa	18	Mati	**Matikanefu
13	Mayano Top Gun	mayano top gun	MAYANO TOP GUN	Mayano Top Gun	Mayano Top Gun Mahasiswa	14	Maya	******Mayano
14	Ryosuke Takahashi	ryosuke takahashi	RYOSUKE TAKAHASHI	Ryosuke Takahashi	Ryosuke Takahashi Mahasiswa	17	Ryos	***Ryosuke T
15	Kadoya Tsukasa	kadoya tsukasa	KADOYA TSUKASA	Kadoya Tsukasa	Kadoya Tsukasa Mahasiswa	14	Kado	******Kadoya

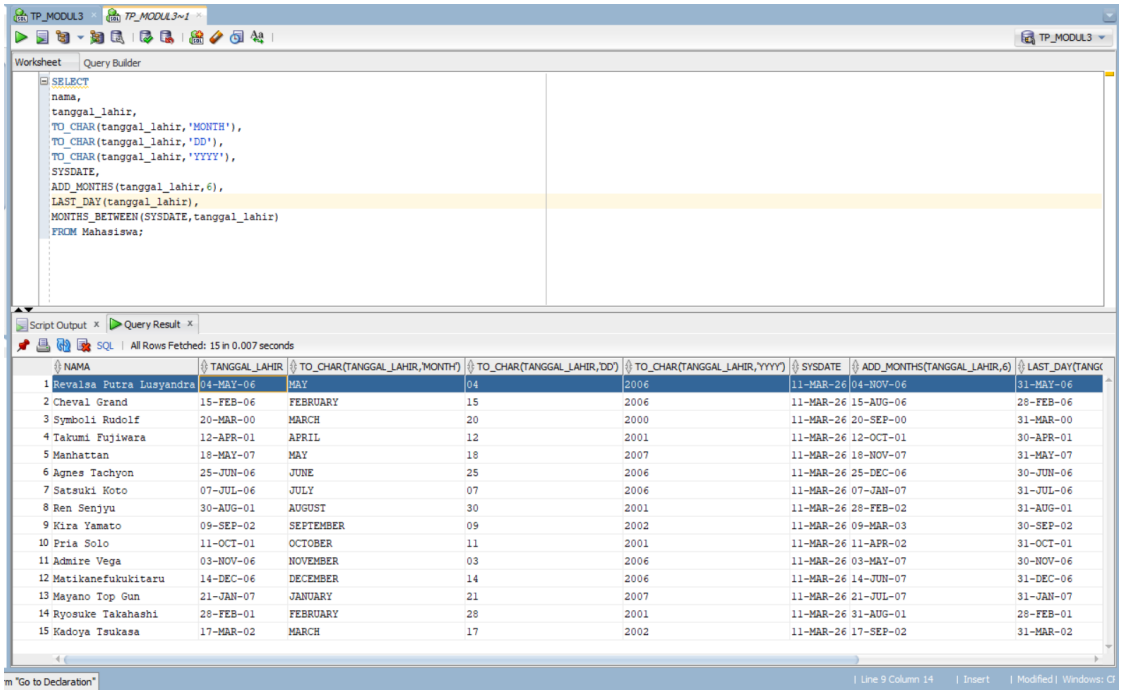


```
SELECT
nama,
LOWER(nama),
UPPER(nama),
INITCAP(nama),
CONCAT(nama,' Mahasiswa'),
LENGTH(nama),
SUBSTR(nama,1,4),
LPAD(nama,20,'*'),
RPAD(nama,20,'*'),
INSTR(email,'@'),
REPLACE(email,'gmail','mail')
FROM Mahasiswa;
```

All Rows Fetched: 15 in 0.007 seconds

	INITCAP(NAMA)	CONCAT(NAMA,'MAHASISWA')	LENGTH(NAMA)	SUBSTR(NAMA,1,4)	LPAD(NAMA,20,'*')	RPAD(NAMA,20,'*')	INSTR(EMAIL,'@')	REPLACE(EMAIL,'GMAIL','MAIL')
1	A Revalsa Putra Lusyandra	Revalsa Putra Lusyandra Mahasiswa	23	Reva	Revalsa Putra Lusyan	Revalsa Putra Lusyan	9	vallreey@mail.com
2	Cheval Grand	Cheval Grand Mahasiswa	12	Chev	********Cheval Grand	Cheval Grand********	7	cheval@mail.com
3	Symboli Rudolf	Symboli Rudolf Mahasiswa	14	Symb	******Symboli Rudolf	Symboli Rudolf******	8	kaichou@mail.com
4	Takumi Fujiwara	Takumi Fujiwara Mahasiswa	15	Taku	*****Takumi Fujiwara	Takumi Fujiwara*****	9	fujiwara@mail.com
5	Manhattan	Manhattan Mahasiswa	9	Manh	**********Manhattan	Manhattan**********	10	manhattan@mail.com
6	Agnes Tachyon	Agnes Tachyon Mahasiswa	13	Agne	*******Agnes Tachyon	Agnes Tachyon*******	8	tachyon@mail.com
7	Satsuki Koto	Satsuki Koto Mahasiswa	12	Sats	********Satsuki Koto	Satsuki Koto********	12	kotosatsuki@mail.com
8	Ren Senjyu	Ren Senjyu Mahasiswa	10	Ren	**********Ren Senjyu	Ren Senjyu**********	10	nexusblue@mail.com
9	Kira Yamato	Kira Yamato Mahasiswa	11	Kira	*********Kira Yamato	Kira Yamato*********	8	freedom@mail.com
10	Pria Solo	Pria Solo Mahasiswa	9	Pria	***********Pria Solo	Pria Solo**********	5	joko@mail.com
11	Admire Vega	Admire Vega Mahasiswa	11	Admi	*********Admire Vega	Admire Vega*********	6	ayabe@mail.com
12	Matikanefukukitaru	Matikanefukukitaru Mahasiswa	18	Mati	**Matikanefukukitaru	Matikanefukukitaru**	7	fukuku@mail.com
13	Mayano Top Gun	Mayano Top Gun Mahasiswa	14	Maya	******Mayano Top Gun	Mayano Top Gun******	5	maya@mail.com
14	Ryosuke Takahashi	Ryosuke Takahashi Mahasiswa	17	Ryos	***Ryosuke Takahashi	Ryosuke Takahashi***	6	rx7fc@mail.com
15	Kadoya Tsukasa	Kadoya Tsukasa Mahasiswa	14	Kado	******Kadoya Tsukasa	Kadoya Tsukasa******	9	magentta@mail.com

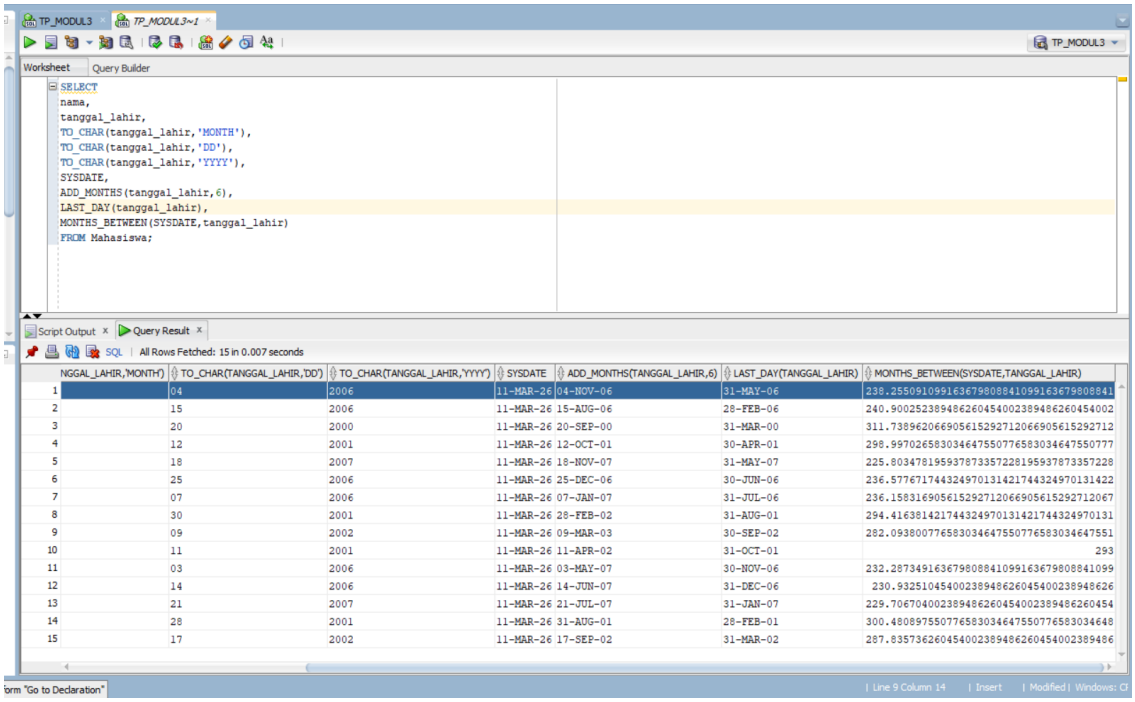
b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH| YEAR]))



```
SELECT
nama,
tanggal_lahir,
TO_CHAR(tanggal_lahir,'MONTH'),
TO_CHAR(tanggal_lahir,'DD'),
TO_CHAR(tanggal_lahir,'YYYY'),
SYSDATE,
ADD_MONTHS(tanggal_lahir,6),
LAST_DAY(tanggal_lahir),
MONTHS_BETWEEN(SYSDATE,tanggal_lahir)
FROM Mahasiswa;
```

All Rows Fetched: 15 in 0.007 seconds

	NAMA	TANGGAL_LAHIR	TO_CHAR(TANGGAL_LAHIR,'MONTH')	TO_CHAR(TANGGAL_LAHIR,'DD')	TO_CHAR(TANGGAL_LAHIR,'YYYY')	SYSDATE	ADD_MONTHS(TANGGAL_LAHIR,6)	LAST_DAY(TANG(
1	Revalsa Putra Lusyandra	04-MAY-06	MAY	04	2006	11-MAR-26	04-NOV-06	31-MAY-06
2	Cheval Grand	15-FEB-06	FEBRUARY	15	2006	11-MAR-26	15-AUG-06	28-FEB-06
3	Symboli Rudolf	20-MAR-00	MARCH	20	2000	11-MAR-26	20-SEP-00	31-MAR-00
4	Takumi Fujiwara	12-APR-01	APRIL	12	2001	11-MAR-26	12-OCT-01	30-APR-01
5	Manhattan	18-MAY-07	MAY	18	2007	11-MAR-26	18-NOV-07	31-MAY-07
6	Agnes Tachyon	25-JUN-06	JUNE	25	2006	11-MAR-26	25-DEC-06	30-JUN-06
7	Satsuki Koto	07-JUL-06	JULY	07	2006	11-MAR-26	07-JAN-07	31-JUL-06
8	Ren Senjyu	30-AUG-01	AUGUST	30	2001	11-MAR-26	28-FEB-02	31-AUG-01
9	Kira Yamato	09-SEP-02	SEPTEMBER	09	2002	11-MAR-26	09-MAR-03	30-SEP-02
10	Pria Solo	11-OCT-01	OCTOBER	11	2001	11-MAR-26	11-APR-02	31-OCT-01
11	Admire Vega	03-NOV-06	NOVEMBER	03	2006	11-MAR-26	03-MAY-07	30-NOV-06
12	Matikanefukukitaru	14-DEC-06	DECEMBER	14	2006	11-MAR-26	14-JUN-07	31-DEC-06
13	Mayano Top Gun	21-JAN-07	JANUARY	21	2007	11-MAR-26	21-JUL-07	31-JAN-07
14	Ryosuke Takahashi	28-FEB-01	FEBRUARY	28	2001	11-MAR-26	31-AUG-01	28-FEB-01
15	Kadoya Tsukasa	17-MAR-02	MARCH	17	2002	11-MAR-26	17-SEP-02	31-MAR-02

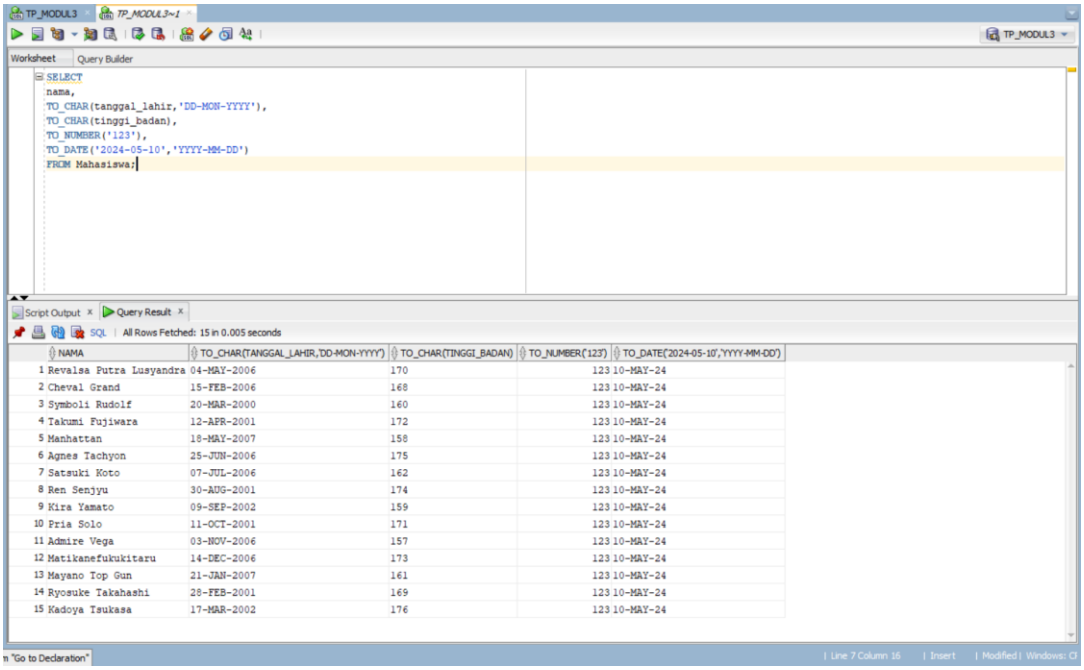


```
SELECT
nama,
tanggal_lahir,
TO_CHAR(tanggal_lahir,'MONTH'),
TO_CHAR(tanggal_lahir,'DD'),
TO_CHAR(tanggal_lahir,'YYYY'),
SYSDATE,
ADD_MONTHS(tanggal_lahir,6),
LAST_DAY(tanggal_lahir),
MONTHS_BETWEEN(SYSDATE,tanggal_lahir)
FROM Mahasiswa;
```

All Rows Fetched: 15 in 0.007 seconds

	NGGAL_LAHIR,'MONTH')	TO_CHAR(TANGGAL_LAHIR,'DD')	TO_CHAR(TANGGAL_LAHIR,'YYYY')	SYSDATE	ADD_MONTHS(TANGGAL_LAHIR,6)	LAST_DAY(TANGGAL_LAHIR)	MONTHS_BETWEEN(SYSDATE,TANGGAL_LAHIR)
1		04	2006	11-MAR-26	04-NOV-06	31-MAY-06	238.2550910991636798088410991636798088410
2		15	2006	11-MAR-26	15-AUG-06	28-FEB-06	240.9002523894862604540023894862604540
3		20	2000	11-MAR-26	20-SEP-00	31-MAR-00	311.7389620669056152927120669056152927
4		12	2001	11-MAR-26	12-OCT-01	30-APR-01	298.9970265830346475507765830346475507
5		18	2007	11-MAR-26	18-NOV-07	31-MAY-07	225.8034781959378733572281959378733572
6		25	2006	11-MAR-26	25-DEC-06	30-JUN-06	236.5776717443249701314217443249701314
7		07	2006	11-MAR-26	07-JAN-07	31-JUL-06	236.1583169056152927120669056152927120
8		30	2001	11-MAR-26	28-FEB-02	31-AUG-01	294.4163814217443249701314217443249701
9		09	2002	11-MAR-26	09-MAR-03	30-SEP-02	282.0938007765830346475507765830346475
10		11	2001	11-MAR-26	11-APR-02	31-OCT-01	293
11		03	2006	11-MAR-26	03-MAY-07	30-NOV-06	232.2873491636798088410991636798088410
12		14	2006	11-MAR-26	14-JUN-07	31-DEC-06	230.9325104540023894862604540023894862
13		21	2007	11-MAR-26	21-JUL-07	31-JAN-07	229.7067040023894862604540023894862604
14		28	2001	11-MAR-26	31-AUG-01	28-FEB-01	300.4808975507765830346475507765830346
15		17	2002	11-MAR-26	17-SEP-02	31-MAR-02	287.8357362604540023894862604540023894

c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))

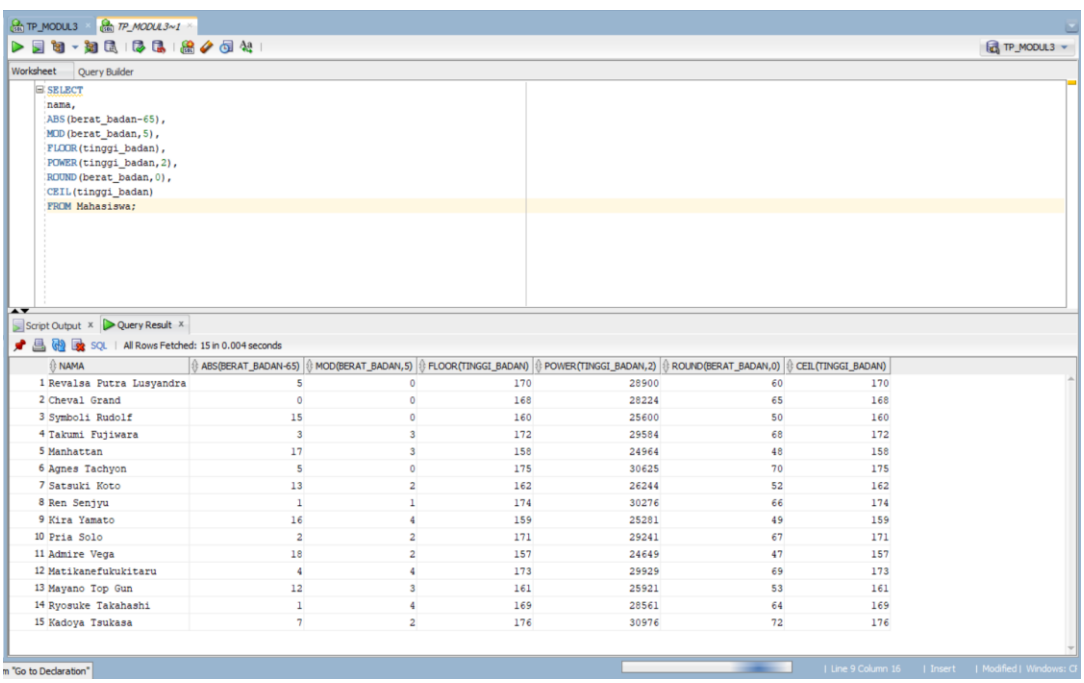


```
SELECT
nama,
TO_CHAR(tanggal_lahir,'DD-MON-YYYY'),
TO_CHAR(tinggi_badan),
TO_NUMBER('123'),
TO_DATE('2024-05-10','YYYY-MM-DD')
FROM Mahasiswa;
```

All Rows Fetched: 15 in 0.005 seconds

	NAMA	TO_CHAR(TANGGAL_LAHIR,'DD-MON-YYYY')	TO_CHAR(TINGGI_BADAN)	TO_NUMBER('123')	TO_DATE('2024-05-10','YYYY-MM-DD')
1	Revalsa Putra Lusyandra	04-MAY-2006	170	123	10-MAY-24
2	Cheval Grand	15-FEB-2006	168	123	10-MAY-24
3	Symboli Rudolf	20-MAR-2000	160	123	10-MAY-24
4	Takumi Fujiwara	12-APR-2001	172	123	10-MAY-24
5	Manhattan	18-MAY-2007	158	123	10-MAY-24
6	Agnes Tachyon	25-JUN-2006	175	123	10-MAY-24
7	Satsuki Koto	07-JUL-2006	162	123	10-MAY-24
8	Ren Senjyu	30-AUG-2001	174	123	10-MAY-24
9	Kira Yamato	09-SEP-2002	159	123	10-MAY-24
10	Pria Solo	11-OCT-2001	171	123	10-MAY-24
11	Admire Vega	03-NOV-2006	157	123	10-MAY-24
12	Matikanefukukitaru	14-DEC-2006	173	123	10-MAY-24
13	Mayano Top Gun	21-JAN-2007	161	123	10-MAY-24
14	Ryosuke Takahashi	28-FEB-2001	169	123	10-MAY-24
15	Kadoya Tsukasa	17-MAR-2002	176	123	10-MAY-24

d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))



```
SELECT
nama,
ABS(berat_badan-65),
MOD(berat_badan,5),
FLOOR(tinggi_badan),
POWER(tinggi_badan,2),
ROUND(berat_badan,0),
CEIL(tinggi_badan)
FROM Mahasiswa;
```

All Rows Fetched: 15 in 0.004 seconds

	NAMA	ABS(BERAT_BADAN-65)	MOD(BERAT_BADAN,5)	FLOOR(TINGGI_BADAN)	POWER(TINGGI_BADAN,2)	ROUND(BERAT_BADAN,0)	CEIL(TINGGI_BADAN)
1	Revalsa Putra Lusyandra	5	0	170	28900	60	170
2	Cheval Grand	0	0	168	28224	65	168
3	Symboli Rudolf	15	0	160	25600	50	160
4	Takumi Fujiwara	3	3	172	29584	68	172
5	Manhattan	17	3	158	24964	48	158
6	Agnes Tachyon	5	0	175	30625	70	175
7	Satsuki Koto	13	2	162	26244	52	162
8	Ren Senjyu	1	1	174	30276	66	174
9	Kira Yamato	16	4	159	25281	49	159
10	Pria Solo	2	2	171	29241	67	171
11	Admire Vega	18	2	157	24649	47	157
12	Matikanefukukitaru	4	4	173	29929	69	173
13	Mayano Top Gun	12	3	161	25921	53	161
14	Ryosuke Takahashi	1	4	169	28561	64	169
15	Kadoya Tsukasa	7	2	176	30976	72	176